

20260727: Standards for joint multi-sample variant calling

- Current best practice for human germline variant calling is **joint genotyping** from per-sample gVCFs, rather than simply merging VCFs.
 - Each sample is processed independently using `HaplotypeCaller -ERC GVCF` .
 - The resulting gVCFs contain genotype likelihoods and reference-confidence information, **not** finalized genotype calls.
 - Joint genotyping uses information from all samples simultaneously to determine:
 - Whether a site is polymorphic.
 - Genotype likelihoods.
 - Allele frequencies.
 - Improved genotype quality, particularly for rare variants.
- Recommended GATK workflow:
 - i. BAM/CRAM
 - ii. `HaplotypeCaller (-ERC GVCF)`

`sample1.g.vcf.gz`
`sample2.g.vcf.gz`
`...`
 - iii. `GenomicsDBImport`
 - iv. `GenotypeGVCFs`
 - v. Multi-sample VCF
 - vi. `VQSR` (or hard filtering)
 - vii. Analysis-ready cohort VCF
- For our cohort:
 - **546 gVCFs (500 Native Hawaiian individuals)** falls well within the recommended range for `GenomicsDBImport` .
 - [Source](#)
 - `CombineGVCFs` is generally intended only for relatively small cohorts and is no longer the preferred scalable solution.
- Example `bash` commands:

```
gatk GenomicsDBImport \
  --sample-name-map cohort.sample_map \
  --genomicsdb-workspace-path cohort_db \
  -L chr1
```

```
gatk GenotypeGVCFs \
  -R reference.fa \
  -V gendb://cohort_db \
  -O cohort.vcf.gz
```

- GATK joint-calling workflow is compatible with downstream tools including PLINK , bcftools , Beagle , SHAPEIT5 , GLIMPSE , Eagle , and **imputation reference panel generation**.
- Alternative pipelines:
 - Solution 1: **DeepVariant + GLnexus**
 - Increasingly common for large whole-genome sequencing projects.
 - Often matches or exceeds GATK SNP/indel accuracy while scaling efficiently.
 - Solution 2: **Illumina DRAGEN**
 - Hardware-accelerated joint variant calling used in many clinical sequencing pipelines.
 - Nonsolution: **bcftools merge**
 - **Not** equivalent to joint genotyping.
 - Only merges existing genotype calls without recomputing genotype likelihoods across individuals.

References

- Van der Auwera GA, Carneiro MO, Hartl C, *et al.* (2013).
 - *From FastQ Data to High-Confidence Variant Calls: The Genome Analysis Toolkit Best Practices Pipeline*.
 - Current Protocols in Bioinformatics 43:11.10.1–11.10.33.
 - DOI: <https://doi.org/10.1002/0471250953.bi1110s43>
 - <https://currentprotocols.onlinelibrary.wiley.com/doi/10.1002/0471250953.bi1110s43>
- Poplin R, Ruano-Rubio V, DePristo MA, *et al.* (2018).
 - *Scaling accurate genetic variant discovery to tens of thousands of samples*.
 - bioRxiv.
 - DOI: <https://doi.org/10.1101/201178>
 - <https://doi.org/10.1101/201178>
- Brouard JS, Schenkel FS, Marete A, *et al.* (2019).

- *The GATK joint genotyping workflow is appropriate for calling variants in RNA-seq experiments.*
- Journal of Animal Science and Biotechnology 10:44.
- DOI: <https://doi.org/10.1186/s40104-019-0359-0>
- <https://jasbsci.biomedcentral.com/articles/10.1186/s40104-019-0359-0>
- Broad Institute GATK guides:
 - [Calling variants on cohorts using HaplotypeCaller in GVCF mode.](#)
 - [The logic of joint calling for germline short variants.](#)

TODO (multi-sample)

- Look for GATK application in a recent sequencing paper

TODO (NH data)

- Remove individual (bug Bryan for information)
- Change EC IDs
- get Bryan's pipeline for m3vcf files
- Generate m3vcf files
 - Compare with previous results to see if there's the same